# Desk Companion

Plan your day by stacking it on your real desk, and see overcommitment before you live it.

Productivity track | New Experience | Target: Best New Productivity Experience

## AT A GLANCE

| | |
|---|---|
| **Track / division** | Productivity (habit tied to morning routine and wind-down) / New Experience |
| **Primary prize** | Best New Productivity Experience ($70,000; runner-up $35,000) |
| **Fallback special award** | Best Agentic Interaction ($25,000) or Best Reason to Come Back ($25,000) |
| **Session length** | Morning ritual about 5 minutes; wind-down about 3 minutes |
| **Room requirement** | A desk or table; tap-to-place fallback on any flat surface |
| **Core tech** | Plane detection and hit-test placement, hand-pinch grabbing, IndexedDB, Claude via a Cloud Run endpoint |
| **Biggest risk** | 'Why put on a headset to plan my day?' |
| **Signature moment** | The stack rising past the red capacity line |
| **Status (2 Oct 2026)** | Full loop built and live; awaiting first physical Quest test |
| **Live build** | desk-companion-1031148889398.us-central1.run.app |
| **Repository** | github.com/4KInc/desk-companion (pull request #1) |

### The one-line pitch

A mixed-reality daily planner where tasks have physical size. Each task becomes a tile whose thickness is its time estimate, and you sort the tiles by hand into zones on your real desk. Stack more than your day can hold and the pile rises past a glowing capacity line: overcommitment you can see and feel.

SECTION 01

# Status report: 2 October 2026

### Where we are in one paragraph

The full Desk Companion loop is built, verified in the IWER emulator from real input, and live on a public HTTPS URL. Every must-ship item from the original scope is done, and most of the nice-to-haves too: focus mode, estimation-bias learning, weekly jar strata, a guided first run and sound design. The build is roughly five weeks ahead of the original schedule with 47 days to the deadline. The one large unknown is the physical headset: nothing has run on a Quest yet, so hand-pinch feel, the flick threshold, dictation support and real frame rate are unverified.

| | |
|---|---|
| **Live build** | https://desk-companion-1031148889398.us-central1.run.app (Cloud Run, us-central1, revision 2) |
| **Code** | github.com/4KInc/desk-companion, branch feat/desk-companion-core, pull request #1 into main (open) |
| **Agent** | Claude Opus 5.5 (low effort, structured output) via a server-side endpoint; key in Google Secret Manager |
| **Deadline** | 18 Nov 2026, 12:00 PT. Plan: submit 17 Nov. 47 days remaining |
| **Biggest open risk** | No physical Quest test yet |

## What is implemented

| Area | Feature | Status | How it was verified |
|---|---|---|---|
| **Placement** | Tray placed on the Space Setup 'table' plane in front of the user, facing them | Done | Emulator, living room and music room |
| **Placement** | Pinch-to-place fallback and a Move tray button | Done | Emulator: reticle followed the ray, pinch placed the tray |
| **Core loop** | Tiles 1 cm thick per 30 min; Now / Later / Drop zones with snapping and auto-stacking | Done | Emulator, real input |
| **Core loop** | Grab by pinch; only the top tile of each pile is grabbable | Done | Emulated controllers; hand pinch untested on device |
| **Core loop** | Capacity line; overflow (judged in minutes) turns line and top tiles red and wobbling | Done | Emulator, real input |
| **Core loop** | Done jar that never resets | Done | Emulator |
| **Agent** | Brain dump: 2D text box or in-XR dictation, split by Claude into titled, estimated, prioritised, categorised tasks | Done | Real Claude calls in dev and production |
| **Agent** | Offline splitter when the agent is unreachable | Done | Emulator and logic check |

| Area | Feature | Status | How it was verified |
|---|---|---|---|
| **Agent** | Capacity nudge: smallest set of lowest-priority tiles to move to Later, Move / Keep | Done | Emulator, both buttons |
| **Agent** | Split a big task: a 2h+ tile landing in Now gets an offer; Claude cracks it into 2 to 4 steps in place | Done | Emulator, real Claude call |
| **Safety** | Undo (8 s) after a flick to tomorrow, a jar drop or a move into Drop; flicks need speed and 12 cm of travel | Done | Emulator: flick and jar both undone |
| **Grab feel** | Hover lift: the tile nearest a hand rises 1.5 cm and glows, so the pinch closes in free air | Done | Emulator: measured 1.5 cm lift |
| **Learning** | Focus mode with a floating timer; plausible focus time becomes the actual | Done | Emulator, real seconds counted |
| **Learning** | 'How long did it take?' card; per-category bias; recap line; adjusted new estimates | Done | Emulator plus logic tests |
| **Daily rhythm** | Flick left to push to tomorrow; Wind down recap; End day, history and streak; rollover | Done | Emulator plus logic tests |
| **Retention** | Weekly strata in the done jar (current week glowing on top) | Done | Logic tests; one layer visible live |
| **Onboarding** | Guided first run: ghost tile, pulsing rings, one-line coach hints | Done | Emulator, all five steps |
| **Polish** | Procedural spatial sound design (no audio files) | Done | Event trace; not listened to on device |
| **Hosting** | Cloud Run server, Secret Manager key, request cap and per-IP rate limit | Live | Anonymous requests to the public URL |
| **Performance** | Two draw groups per tile, room meshes off | Done | 102 draw calls with 13 tiles (was 206) |
| **Persistence** | Tray position remembered across sessions with anchors | Not built | Tray re-detects the desk each session |
| **Store** | PWA manifest, icons, Quest APK packaging | Not started | Optional; the submission only needs a URL |
| **Accessibility** | Large-tile and high-contrast options | Not started | Planned |

## Verification so far

- **Logic tests:** four scripts covering day rollover and streaks (including month boundaries), weekly jar layers, the estimation-bias model and the coach steps. All pass.
- **Live emulator runs:** every feature above was exercised through real XR input (ray clicks, squeeze-grab, flick swings, chest release) with named ECS and UI state checked after each step.
- **Production:** page, assets, MIME types, caching, 404 for secrets, 413 for oversized requests, 429 after 20 requests in 10 minutes, and live Claude calls from the public URL.

- **Performance (desktop emulator):** 102 draw calls and 23k triangles with 13 tiles; median frame 8.3 ms with no dropped frames; the app's own systems cost about 0.09 ms per frame.
- **Not yet verified:** anything on a physical Quest.

## What changed from the original plan

| Original plan | What was built | Why |
|---|---|---|
| **Host on Vercel** | Google Cloud Run with Secret Manager | The developer chose GCP; one container serves the app and the API. |
| **A fast, cheap model** | Claude Opus 5.5 at low effort with structured output | Best quality at acceptable latency (2 to 5 s per brain dump). |
| **Free-form stacking** | Snap zones, auto-stacking, top-of-pile grabbing | Thin tiles millimetres apart made proximity grab pick the wrong tile. |
| **Agent writes the nudge** | Deterministic nudge from priorities and minutes | Fast, offline-safe and never claims knowledge it lacks; Claude supplies priorities and categories. |
| **Focus mode and bias learning as nice-to-haves** | Both built | The schedule ran far ahead; they power the Reason to Come Back story. |
| **Anchors keep the tray on the desk** | Desk re-detected each session, plus Move tray | Persistent anchors in Quest Browser are unverified; re-detection takes about 2 seconds. |
| **Room mesh detection** | Turned off | Unused by the app; it doubled draw calls. |

## What is next

| Dates | Focus | Done when |
|---|---|---|
| **2 - 8 Oct** | **Quest smoke test** on the live URL: hand pinch on tiles, flick threshold, chest-release radius, label legibility, dictation support, real frame rate with 15+ tiles | Every interaction works by hand on device; issues logged and fixed |
| **9 - 15 Oct** | Device fixes; persistent anchors if Quest Browser supports them; large-tile and high-contrast options | Tray reappears on the desk without re-placing (or documented fallback) |
| **16 - 22 Oct** | Use it daily for a week (real bias data); art pass on tiles and jar; hover and grab highlights for hands; sound mix on device | A week of real use with no blocking bugs |
| **23 Oct - 5 Nov** | Playtests with 5+ people in their own rooms; first-five-minutes iteration; **start recording video footage 26 Oct** | Cold testers finish the morning ritual unaided |
| **6 - 12 Nov** | Edit the under-3-minute video; final submission text; optional PWA manifest and icons | Video and text ready |
| **13 - 17 Nov** | Freeze, final deploy, verify the public URL, submit on 17 Nov | Submitted a day early |

### Open decisions and housekeeping

- **Endpoint hardened (3 Oct):** rate limit keyed on the real client IP (last X-Forwarded-For entry), same-origin check on POSTs, a global daily cap, and no task text in logs.
- **Rotate the Anthropic API key:** the current key was pasted into a chat. Create a new one, update .env and the Secret Manager secret, then redeploy.
- **Merge pull request #1** into main once reviewed.
- **gcloud account:** the developer's owner account is currently active; switch back to the service account when appropriate.
- **Cost:** Cloud Run scales to zero with at most 2 instances; the brain-dump endpoint is limited to 20 requests per visitor per 10 minutes.
- **Store packaging:** decide whether to ship a Horizon Store PWA in addition to the URL.

SECTION 02

## The idea

To-do apps are lists, and lists hide the thing that actually hurts: you planned nine hours of work into a five-hour day. In mixed reality, tasks can have physical size. Put too much on your desk and you see the pile overflow.

Desk Companion is not a to-do list in VR. It is your workload made physical, sitting on your actual desk. It is designed around two short, recurring moments: a 5-minute morning plan and a 3-minute evening wind-down.

> **The core mechanic**
>
> Every task is a tile with real thickness: a 30-minute task is a thin tile, a 3-hour task a thick block (for example 1 cm per 30 minutes, so an 8-hour day is a 16 cm stack). A translucent capacity line at the Now zone is set from the hours you actually have today. Stack past it and the top tiles glow red and wobble. You feel the overcommitment instead of reading about it.

SECTION 03

## How a day works

### Morning ritual (about 5 minutes)

1. **The tray appears.** Put on the headset at your desk. The app finds the table surface and places a soft glowing day tray on it, anchored to your real desk.
2. **Brain dump.** Dictate or type what is on your mind. The agent splits it into task tiles, each with a time estimate. Tiles drop onto the desk.
3. **Sort by hand.** Pinch-grab tiles into three zones drawn on the desk: Now, Later and Drop. Tiles snap into place and stack automatically.
4. **Capacity line.** As the Now stack grows it approaches the line set by today's available hours. Past it, the top tiles glow red and wobble.
5. **Agent nudge.** A short suggestion appears beside the stack, such as 'Now is 1.5 h over. These two are lowest priority. Move them to Later?' One pinch accepts.

### During the day (optional)

Pull a tile toward your chest and it becomes a focus card, with a timer floating above the desk while the other tiles dim.

### Wind-down (about 3 minutes)

- Sweep finished tiles into a done jar on the corner of your desk. The jar fills across days and weeks and never resets.
- For unfinished tiles, flick left to push them to tomorrow, or drop them in Drop.
- For each finished tile, a quick slider asks roughly how long it took (optional, one pinch).
- The agent gives a one-line recap: 'You planned 6 h and did 4.5 h. Your estimates on writing tasks run about 40% low.'

SECTION 04

## Improvements over the first draft of this concept

| Original plan | Change | Why |
|---|---|---|
| **Free-form stacking by hand** | Tiles snap into zones and stack automatically; the pinch only chooses the zone. | Precise stacking with hand tracking is fiddly and a failed drop ruins the demo. |
| **Agent notices dependencies like 'waiting on a colleague's review'** | Nudges only use information the user actually gave: estimates, priority, capacity, and explicitly stated dependencies. | Judges will test the agent; it should never claim knowledge it cannot have. |
| **Estimation-bias learning as a headline feature** | Show it in the video as a feature that grows over weeks; do not headline it. | It needs weeks of real data that will not exist by the deadline. |
| **Voice input assumed** | Voice if available; otherwise a companion phone page (scan a code) or quick-pick templates. | Speech recognition inside a WebXR session in Quest Browser is unverified. |
| **Plane detection assumed** | Tap-to-place on any flat surface is a first-class path. | A table is only found if the user labelled it during Space Setup. |

SECTION 05

## Why passthrough and the real room matter

- **The content lives on your desk.** The tray, zones and stacks sit on your real work surface, at real scale. With passthrough off the concept breaks, which is what the judges want.
- **Physical size carries meaning.** Height on your desk is a real measurement you can compare with the objects around you: 'today's plan is taller than my coffee mug' is an intuitive signal.
- **The done jar is a permanent object in your room.** It sits on your desk corner every day, filling up.

SECTION 06

## The first five minutes

| Time | What happens |
|---|---|
| **0:00** | Passthrough. A soft light traces the edge of the desk and a day tray settles onto it. |
| **0:20** | A prompt: 'What is on your mind today?' The user speaks or uses the fallback input. |
| **1:00** | Tiles drop onto the tray one by one, each labelled with a title and time estimate. |
| **1:30** | Three zones glow: Now, Later, Drop. The first tile pulses; the user pinches it and drops it in Now. It snaps and stacks. |
| **3:00** | The Now stack approaches the capacity line; the line brightens. One more tile and it glows red and wobbles. |
| **3:30** | The agent nudge appears; one pinch moves two tiles to Later and the stack settles under the line with a satisfying sound. |
| **4:30** | Done. 'See you this evening.' The first run is the tutorial, done with the user's own tasks. |

SECTION 07

## The agent

The AI makes the experience faster, but the experience stands without it, which satisfies the take-it-away test: tiles can be created manually, sorted physically, checked against capacity, and swept into the jar with no AI at all.

| Agent job | What it does | Visible as | Status |
|---|---|---|---|
| **Split the brain dump** | Claude turns free text into tasks with a title, estimate, priority and category | Tiles dropping into the Inbox | Built |
| **Capacity nudge** | Picks the smallest set of lowest-priority tiles that fixes an overflow | A card on the panel with Move / Keep | Built (deterministic, uses Claude's priorities) |
| **Evening recap** | Planned versus finished, tomorrow pile, dropped, carried over, bias line | The Wind down card | Built (local template) |

| Agent job | What it does | Visible as | Status |
|---|---|---|---|
| **Estimation bias** | Per-category actual/estimate ratio from the last 20 tasks | Recap line and adjusted new estimates | Built |
| **Split a big task** | Offered when a 2h+ tile lands in Now; Claude names 2 to 4 concrete steps | The tile cracks and its steps fly into the pile | Built (offline: equal parts) |

### Implementation

- A server-side endpoint (POST /api/brain-dump on the Cloud Run service, shared with the Vite dev server) holds the API key so it never reaches the headset.
- It calls Claude Opus 5.5 at low effort with structured output (a fixed schema), with server-side refusal fallbacks, so the app always receives valid task data. Typical latency is 2 to 5 seconds.
- The app sends only the text needed for the request and stores all tasks locally in IndexedDB. No accounts and no server-side storage.
- If the network or AI is unavailable, the app falls back to manual tile creation with preset durations (15 min, 30 min, 1 h, 2 h).

SECTION 08

## Progression and reasons to come back

| System | Description |
|---|---|
| **The done jar** | Fills with finished tiles across days and weeks and never resets. The main visible growth. |
| **Streaks** | Consecutive days with both a morning plan and a wind-down. |
| **Honest estimates** | Over weeks, the app learns your estimation bias per category and shows how your planning improves. |
| **Weekly shelf** | At the end of each week, the jar's contents compress into a labelled layer, so months of work are visible as strata. |

SECTION 09

## Accessibility

- Voice input where available, with typing and preset alternatives.
- Large tiles and generous snap zones for limited fine motor control.
- Capacity warnings use motion, height and pattern, not only red colour.
- Captions for all audio; high-contrast mode for tile text.

SECTION 10

## Art direction and sound

Soft, tactile and calm: frosted-glass tiles with clear typography, gentle glows for zones, a warm glass jar. The overflow moment is the one dramatic beat: red glow, wobble and a low hum. Sounds are soft clicks for snapping, a glassy chime for tiles entering the jar.

SECTION 11

## Technical architecture (IWSDK)

| Feature | Used for |
|---|---|
| **Plane detection** | Finding the desk (the scaffold already enables it). |
| **Hit test** | Tap any surface to place the tray when no table is found. |
| **Anchors** | Keeping the tray and jar in place across days (persistence in Quest Browser needs checking). |
| **Grabbing and hand tracking** | Picking up and dropping tiles; flick and sweep gestures. |
| **UI (UIKit)** | Tile text, nudge cards, recap text. |
| **IndexedDB** | All task data, history and jar contents. |
| **Serverless function** | The agent calls. |

| System | Responsibility |
|---|---|
| **DeskSetupSystem** | Places the tray and jar; saves the anchors. |
| **TaskTileSystem** | Creates tiles from task data; sets thickness from the estimate. |
| **ZoneSnapSystem** | Snaps dropped tiles into zones and restacks them. |
| **CapacitySystem** | Compares stack height with the capacity line; triggers the overflow effects. |
| **AgentSystem** | Calls the serverless function and turns responses into tiles and nudges. |
| **FocusSystem** | The focus card and timer. |
| **JarSystem** | Done jar, streaks and weekly strata. |

SECTION 12

## Risks and mitigations

| Risk | Severity | Mitigation |
|---|---|---|
| **'Why a headset to plan my day?'** | High | Target people who already work in a headset; make the overflow moment striking in the first 20 seconds of the video. |
| **Productivity is a crowded idea** | High | Lead with the capacity line and the done jar, not feature breadth. |
| **Voice input unavailable in WebXR** | Medium | Phone companion page or presets; test in week 1. |
| **Fiddly grabbing** | Medium | Snap zones and automatic stacking. |
| **Anchor persistence unreliable** | Medium | Re-detect the desk each session; store jar contents in IndexedDB regardless. |
| **Agent overreaches** | Medium | Structured output; nudges based only on known data. |

### Week-1 checks: status

| Question | Answer so far |
|---|---|
| **Does the Web Speech API work inside a WebXR session in Quest Browser?** | Works in the desktop emulator; unverified on Quest. The Dictate button hides itself when unsupported, and the 2D brain-dump box is the fallback. |
| **Do persistent anchors survive closing and reopening the app?** | Not tested; not used. The tray re-detects the desk each session, and Move tray re-places it by hand. |
| **Does plane detection return a labelled table in your own room?** | Yes in two emulated rooms; real rooms unverified. Pinch-to-place covers rooms without a labelled table. |
| **Can hand tracking reliably pinch a 6 cm tile lying on a desk?** | Unverified on device. Tiles are 10 x 7 cm, pinch-to-grab is enabled, and only pile tops are grabbable. |

SECTION 13

## Original seven-week build plan, with progress

The original plan is kept for reference. Content planned for weeks 2 to 5 was finished in the first week, in the emulator; the remaining weeks shift to device testing, playtesting and the video (see 'What is next' above).

| Week | Original goal | Status on 2 Oct |
|---|---|---|
| **1** | Platform checks (speech, anchors, planes, pinch); tray placement | Placement done; device checks pending |
| **2** | Tiles with thickness; zones; snapping and stacking | Done |
| **3** | Capacity line and overflow; agent function for brain dump | Done |
| **4** | Agent nudge; first-run flow; IndexedDB persistence | Done |

| Week | Original goal | Status on 2 Oct |
|---|---|---|
| **5** | Wind-down: done jar, flick to tomorrow, recap; focus card | Done |
| **6** | Performance; playtests over several days; accessibility options | Performance pass done (desktop); playtests and accessibility next |
| **7** | Polish; record video; write submission | Not started |
| **Ship** | Final checks; submit on 17 Nov | Planned |

SECTION 14

## Scope

| Must ship | Nice to have | Cut |
|---|---|---|
| **Desk detection and tray; tiles; three zones; capacity line; done jar; agent brain dump. All built.** | Focus mode; estimation-bias learning; streak; weekly strata; sound design. **All built.** Persistent anchors and accessibility options still open. | Calendar integrations; accounts; sync across devices. Still cut. |

SECTION 15

## Demo video plan (under 3 minutes)

| Time | Beat | Shot |
|---|---|---|
| **0:00 - 0:15** | Hook | Tiles piling past the red line on a real desk, wobbling. |
| **0:15 - 1:30** | Morning ritual | Real-time brain dump, tiles dropping, sorting by hand. |
| **1:30 - 2:00** | The nudge | The agent's suggestion; one pinch; the stack settles below the line. |
| **2:00 - 2:40** | Wind-down | Sweeping tiles into the jar; flicking unfinished ones to tomorrow. |
| **2:40 - 2:55** | Coming back | Montage of the jar filling across days. |

SECTION 16

## Submission draft

| | |
|---|---|
| **Name** | Desk Companion |
| **Tagline (under 140 characters)** | Plan your day by stacking it on your real desk. See overcommitment before you live it. |
| **Track / division** | Productivity / New Experience |
| **Target launch** | Web release Q1 2027 |

### Description (first draft)

To-do lists hide the problem that hurts most: planning more than a day can hold. Desk Companion gives every task physical size on your real desk.

Each morning, tell Desk Companion what is on your mind. An AI agent turns it into task tiles whose thickness is their time estimate. Sort them by hand into Now, Later and Drop. When the Now stack rises past your capacity line, the top tiles glow red and wobble, and the agent suggests what to move. Each evening, sweep finished tiles into a done jar on your desk that fills up over weeks.

We built it with the Immersive Web SDK: plane detection finds your desk (with a pinch-to-place fallback for any surface), hand-pinch grabbing with a hover lift picks up tiles, and Claude runs behind a small server-side endpoint so the key never reaches the headset. The agent splits your brain dump into estimated tasks, offers to crack big tasks into smaller steps, and learns how long your work really takes. Everything also works without it: you can create, sort and finish tasks entirely by hand.

SECTION 17

## How it maps to the judging criteria

| Criterion (25% each) | Evidence in Desk Companion |
|---|---|
| **Innovation and Creativity** | Workload as physical height; overcommitment made visible and tangible. |
| **Experience Design** | Two recurring rituals; first run is the tutorial; content sits low and central. |
| **Technical Implementation** | Plane detection with fallback, anchors, grabbing, an AI agent with structured output. |
| **Polish and Presentation** | The overflow shot is a strong hook; calm, tactile visuals. |

### Hard requirement and design guidelines

| Test | What the rules ask | How this idea passes |
|---|---|---|
| **Hands-first (hard requirement)** | Complete the entire experience without ever pairing a controller. | Every action is a pinch, grab, sweep or flick. |
| **Airplane seat test** | Every interaction works within a two-foot radius, seated. | Everything happens on the desk within arm's reach. |
| **One bus stop test** | Fast cold start, clean pause/resume, a satisfying moment in 10 minutes or less. | Morning is 5 minutes, wind-down 3 minutes; each is complete on its own. |
| **Take it away test** | Remove any third-party service and a project still stands. | Remove the AI and you can still create tiles, sort them, see capacity and fill the jar. |

### Competition rules that shape this plan

- **One prize per entry.** Rule 7 limits each entry to a single prize, so this brief aims at one track prize and names one special award as a fallback rather than trying to stack awards.
- **Submission:** a public HTTPS link (GitHub Pages, Vercel or similar) that stays live until the winner announcement (about 11 Dec 2026), a public YouTube/Vimeo video under 3 minutes, and the Devpost form (name, 140-character tagline, track, division, description of about 500 words, hand-interaction notes, target launch date).
- **No changes after the deadline.** Judging uses the build as it stood at 18 Nov 2026, 12:00 PT. Plan to submit on 17 Nov and freeze the hosted build.
- **Content rules:** English (or English subtitles), no identifiable real people other than entrants, no brand names or logos, all music, art and fonts properly licensed or original.
- **Video rules:** real footage from a Quest or the XR simulator; do not lean on AI-generated video to carry the pitch.
- **Performance floor:** judges expect at least 60 fps on Quest hardware. Target 72 fps.
- **Wrapper rule:** the AI is an accelerator, not the product. Make sure the video shows manual tile creation too.

SECTION 18

## Honest weaknesses

- The biggest risk is the question 'why put on a headset to plan my day?'
- Productivity apps are a common idea; others will build task managers too.
- Voice input in Quest Browser needs checking.
- No multiplayer, so the Social award is out of reach (which is fine).

SECTION 19

## Scorecard and verdict

| Dimension | Score / 10 | Comment |
|---|---|---|
| **Originality** | 7 | The capacity line is original; the category is not. |
| **Only possible in MR** | 8 | Physical size on a real desk is the point. |
| **Works in untested rooms** | 8 | One surface plus a fallback. |
| **Solo feasibility in 7 weeks** | 7 | Moderate; the agent adds an external dependency. |
| **Prize ceiling** | $70k | Productivity track; Agentic Interaction as a fallback. |

**Verdict**

A solid, credible Productivity entry built around one strong mechanic. Choose it if you prefer the Productivity track or want to compete for the Agentic Interaction award; focus everything on the overflow moment and the done jar.